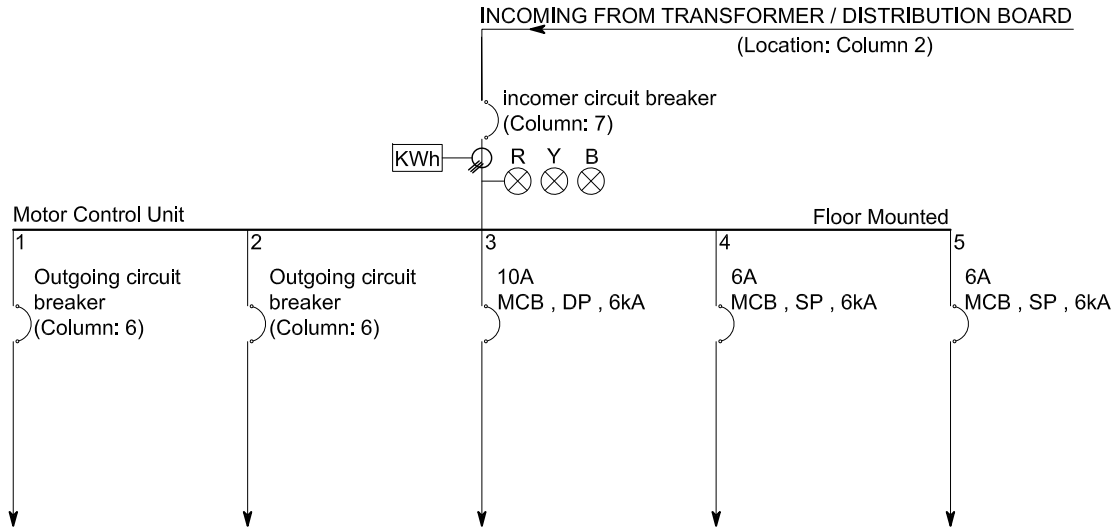


Note.
1- PLEASE REFER TO BELOW TABLE FOR THE COLUMN NUMBERS MENTIONED IN THIS DRAWING



Equipment Description	VENTILLATION FAN 1	VENTILATION FAN 2 (if Applicable)	I&C System	Lights	Spare/misc.
Equipment Rating (kW))	COLUMN: 3	COLUMN: 3	--	--	--
Quantity (Nos.)	COLUMN: 4	COLUMN: 4	02	04	04
Starter Type	COLUMN: 5	COLUMN: 5	--	--	--

Column:1	Column:2	Column:3	Column:4	Column:5	Column:6	Column:7
Drop Structure Odor Control System						
Sr.No.	Location	installed Equipment Details				Incoming Circuit Breaker
		Equipment Rating	Installed Qty.	Starter Type	Feeder Circuit Breaker	
		(kW)	(Nos.)			
1	Main Mir (D1)	1.5	1	DOL	MCB -TP, 6A-16kA	MCCB-TP, 20A - 25kA
2	Infantry (D2)	4	1	DOL	MCB -TP, 10A-16kA	MCCB-TP, 20A - 25kA
3	Bird Market (D3)	7.5	1	DOL	MCB -TP, 20A-16kA	MCCB-TP, 32A - 25kA
		1.5	1	DOL	MCB -TP, 6A-16kA	
4	Gov House (D4)	1.5	1	DOL	MCB -TP, 6A-16kA	MCCB-TP, 20A - 25kA
5	Lakshami Chowk (D5)	5.5	1	DOL	MCB -TP, 16A-16kA	MCCB-TP, 32A - 25kA
6	Line-A Qurtaba (D6)	15	2	DOL	MCB -TP, 40A-16kA	MCCB-TP, 50A - 25kA
7	Grumangat(D7)	1.5	1	DOL	MCB -TP, 6A-16kA	MCCB-TP, 20A - 25kA
8	Q Block (D8)	1.5	1	DOL	MCB -TP, 6A-16kA	MCCB-TP, 20A - 25kA
9	G Block (D10)	5.5	1	DOL	MCB -TP, 16A-16kA	MCCB-TP, 32A - 25kA
		2.2	1	DOL	MCB -TP, 6A-16kA	
10	Zafar Ali Rd (D11)	1.5	1	DOL	MCB -TP, 6A-16kA	MCCB-TP, 20A - 25kA
11	Ichra (D12)	5.5	1	DOL	MCB -TP, 16A-16kA	MCCB-TP, 32A - 25kA
		1.5	1	DOL	MCB -TP, 6A-16kA	
12	Shama (D14)	5.5	1	DOL	MCB -TP, 16A-16kA	MCCB-TP, 32A - 25kA
		1.5	1	DOL	MCB -TP, 6A-16kA	
13	Line-B Qurtaba (D16)	11	1	ASD	MCB -TP, 32A-16kA	MCCB-TP, 50A - 25kA
		7.5	1	DOL	MCB -TP, 20A-16kA	
14	Lyton (D17)	1.5	1	DOL	MCB -TP, 6A-16kA	MCCB-TP, 20A - 25kA
15	PIA (D18)	1.5	1	DOL	MCB -TP, 6A-16kA	MCCB-TP, 20A - 25kA
16	Poonch (D19)	4	1	DOL	MCB -TP, 16-16kA	MCCB-TP, 20A - 25kA

SYMBOL	LEGENDS
	DESCRIPTION
	CIRCUIT BREAKER
MCCB	MOLDED CASE CIRCUIT BREAKER
MCB	MINIATURE CIRCUIT BREAKER
FP	FOUR POLE
TP	THREE POLE
DP	DOUBLE POLE
SP	SINGLE POLE
	EARTH SWITCH
	AMMETER SELECTOR SWITCH
	VOLTMETER SELECTOR SWITCH
	DIGITAL POWER METER
	CURRENT TRANSFORMER
	INDICATION LIGHT

(SHEET 1 OF 1)

LAHORE WATER & WASTEWATER MANAGEMENT PROJECT - SEWERAGE SYSTEM FROM LARECH COLONY TO GULSHAN-E-RAVI, LAHORE (THROUGH TRENCHLESS TECHNOLOGY)



WATER AND SANITATION
AGENCY, LAHORE



A Joint Venture

04		
03		
02		
01		
REV.	DATE	DESCRIPTION

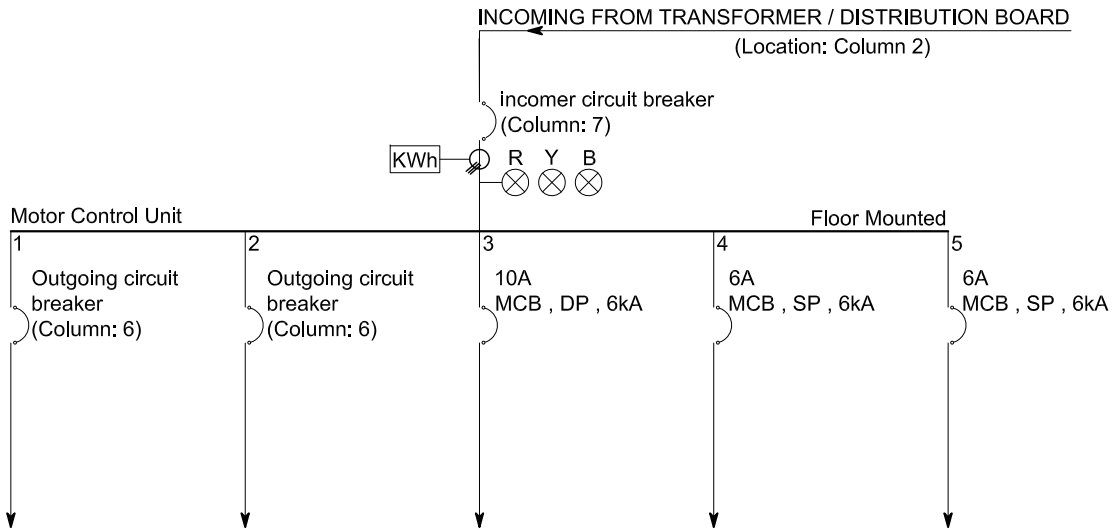
DRAWN	MUHAMMAD ARSLAN
CHECKED BY	SHAHZAIB SHABBIR
RECOMMENDED	
CHD./VER.	
APPROVED	

ODOR CONTROL SYSTEM (SINGLE LINE DIAGRAM) TYPICAL

DATE:-
SEPTEMBER, 2024

DRAWING NO.:-
LW-TD-151

Note.
1- PLEASE REFER TO BELOW TABLE FOR
THE COLUMN NUMBERS MENTIONED IN
THIS DRAWING



Equipment Description	SCREEN	CONVEYOR (if Applicable)	I&C System	Lights	Spare/misc.
Equipment Rating (kW))	COLUMN: 3	COLUMN: 3	--	--	--
Quantity (Nos.)	COLUMN: 4	COLUMN: 4	02	04	04
Starter Type	COLUMN: 5	COLUMN: 5	--	--	--

Column:1	Column:2	Column:3	Column:4	Column:5	Column:6	Column:7
Screens Area						
Sr.No.	Location	installed Equipment Details				Incoming Circuit Breaker
		Equipment Rating	Installed Qty.	Starter Type	Feeder Circuit Breaker	
		(kW)	(Nos.)			
1	Main Mir	1.1	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
2	infantry	3	2	DOL	MCB -TP, 10A-16kA	MCB -TP, 16A-16kA
3	Bird Market	1.1	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
4	Larech Colony	0.37	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
5	Ghazi Mohallah	0.37	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
6	Mohammad Nagar	0.37	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
7	Lakshmi Chowk	1.5	2	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
	Lakshmi Chowk Belt Conveyor	0.75	1	DOL	MCB -TP, 6A-16kA	
8	Gov House	0.75	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
9	Janki Devi	0.37	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
10	Grumangat	1.1	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
11	Q Block	1.1	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
12	Center Point	1.5	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
13	G Block	1.5	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
14	Shareef Colony	0.55	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
15	Zafar Ali Rd	0.37	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
16	Shadman	2.2	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
17	Shama	0.75	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
18	Ichra	1.1	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
19	Rasool Park	2.2	2	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
	Rasool Park Belt Conveyor	0.75	1	DOL	MCB -TP, 6A-16kA	
20	Lyton /Lake Rd	1.1	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
21	PIA	1.1	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
22	Poonch	1.5	2	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
	Poonch Belt Conveyor	0.75	1	DOL	MCB -TP, 6A-16kA	

(SHEET 1 OF 1)

LAHORE WATER & WASTEWATER MANAGEMENT PROJECT - SEWERAGE SYSTEM FROM LARECH COLONY TO GULSHAN-E-RAVI, LAHORE (THROUGH TRENCHLESS TECHNOLOGY)



WATER AND SANITATION
AGENCY, LAHORE



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04		
03		
02		
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REV.	DATE	DESCRIPTION

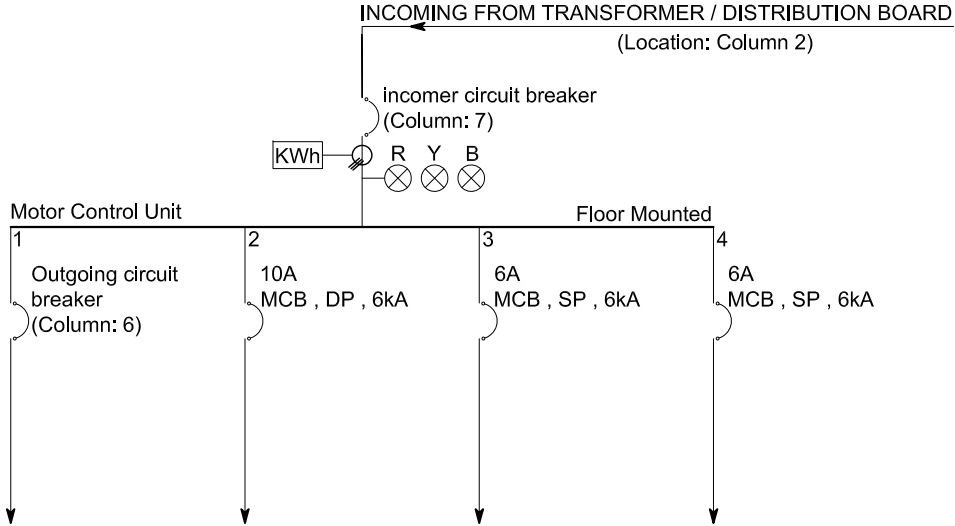
DRAWN	MUHAMMAD ARSLAN
CHECKED BY	SHAHZAIB SHABBIR
RECOMMENDED	
CHD./VER.	
APPROVED	

SCREENS (SINGLE LINE DIAGRAM) TYPICAL

DATE:-
SEPTEMBER, 2024

DRAWING NO.:-
LW-TD-152

Note.
1- PLEASE REFER TO BELOW TABLE FOR THE COLUMN NUMBERS MENTIONED IN THIS DRAWING



Equipment Description	SCREEN	I&C System	Lights	Spare/misc.
Equipment Rating (kW)	COLUMN: 3	--	--	--
Quantity (Nos.)	COLUMN: 4	02	04	04
Starter Type	COLUMN: 5	--	--	--

Column:1	Column:2	Column:3	Column:4	Column:5	Column:6	Column:7
Ventillation System						
Sr.No.	Location	installed Equipment Details				Incoming Circuit Breaker
		Equipment Rating	Installed Qty.	Starter Type	Feeder Circuit Breaker	
		(kW)	(Nos.)			
1	E1	5.5	1	DOL	MCB -TP, 16A-16kA	MCB -TP, 32A-16kA
2	E2	5.5	1	DOL	MCB -TP, 16A-16kA	MCB -TP, 32A-16kA
3	E3	7.5	1	DOL	MCB -TP, 20A-16kA	MCB -TP, 40A-16kA
4	E4	7.5	1	DOL	MCB -TP, 20A-16kA	MCB -TP, 40A-16kA
5	E5	7.5	1	DOL	MCB -TP, 20A-16kA	MCB -TP, 40A-16kA
6	E6	2.2	2	DOL	MCB -TP, 10A-16kA	MCB -TP, 32A-16kA
7	E7	3	1	DOL	MCB -TP, 10A-16kA	MCB -TP, 32A-16kA
8	E8	5.5	1	DOL	MCB -TP, 16A-16kA	MCB -TP, 32A-16kA
9	E9	5.5	1	DOL	MCB -TP, 16A-16kA	MCB -TP, 32A-16kA
10	E10	7.5	1	DOL	MCB -TP, 20A-16kA	MCB -TP, 40A-16kA
11	E11	22	1	ASD	MCB -TP, 50A-16kA	MCB -TP, 63A-16kA
12	E12	22	1	ASD	MCB -TP, 50A-16kA	MCB -TP, 63A-16kA
13	E13	15	1	ASD	MCB -TP, 40A-16kA	MCB -TP, 63A-16kA

	LEGENDS
SYMBOL	DESCRIPTION
	CIRCUIT BREAKER
MCCB	MOLDED CASE CIRCUIT BREAKER
MCB	MINIATURE CIRCUIT BREAKER
FP	FOUR POLE
TP	THREE POLE
DP	DOUBLE POLE
SP	SINGLE POLE
	EARTH SWITCH
	AMMETER SELECTOR SWITCH
	VOLTMETER SELECTOR SWITCH
	DIGITAL POWER METER
	CURRENT TRANSFORMER
	INDICATION LIGHT

(SHEET 1 OF 1)

LAHORE WATER & WASTEWATER MANAGEMENT PROJECT - SEWERAGE SYSTEM FROM LARECH COLONY TO GULSHAN-E-RAVI, LAHORE (THROUGH TRENCHLESS TECHNOLOGY)



WATER AND SANITATION
AGENCY, LAHORE



A Joint Venture

04		
03		
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REV.	DATE	DESCRIPTION

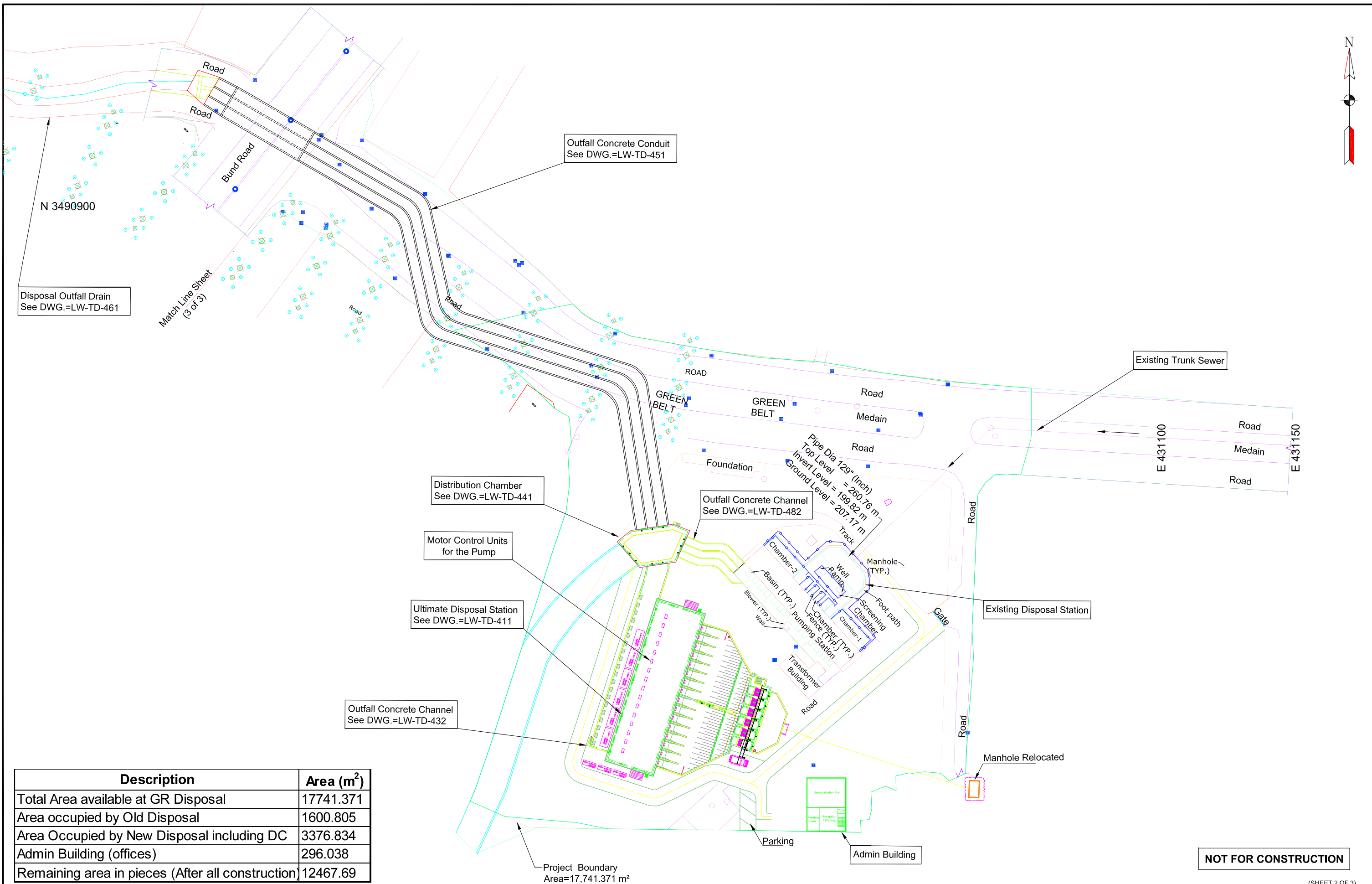
DRAWN	MUHAMMAD ARSLAN
CHECKED BY	SHAHZAIB SHABBIR
RECOMMENDED	
CHD./VER.	
APPROVED	

VENTILATION SYSTEM (SINGLE LINE DIAGRAM) TYPICAL

DATE:-
SEPTEMBER, 2024

DRAWING NO.:-
LW-TD-153

Annexure-B



Description	Area (m ²)
Total Area available at GR Disposal	17741.371
Area occupied by Old Disposal	1600.805
Area Occupied by New Disposal including DC	3376.834
Admin Building (offices)	296.038
Remaining area in pieces (After all construction)	12467.69

(SHEET 2 OF 3)

LAHORE WASTEWATER AND DRAINAGE MANAGEMENT PROJECT (LWDMP) - SEWERAGE SYSTEM FROM LARECHS COLONY TO GULSHAN-E-RAVI, LAHORE (THROUGH TRENCHLESS TECHNOLOGY)



**WATER AND SANITATION
AGENCY, LAHORE**



A Joint Venture

04			DRAWN	MUHAMMAD HAMMAD NASEER
03			CHECKED BY	MUHAMMAD SAQIB RAZA
02			RECOMMENDED	
01			CHD./VER.	
REV.	DATE	DESCRIPTION	APPROVED	

	MUHAMMAD HAMMAD NASEER
	MUHAMMAD SAQIB RAZA

ULTIMATE DISPOSAL STATION MASTER LAYOUT PLAN PUMP HOUSE		
DATE:-	NOVEMBER, 2025	DRAWING NO.:-
		LW - TD - 402

Annexure-C

Sr No	Name of Lift Station	Number & Capacity of Pumps (Motor Rating)	Transformer (KVA)	Generator (KVA)	Remarks
1	Mian Mir	1 No x 4 Cusec (60 HP)	100	Not Available	
2	Infantry Road	1 No x 4 Cusec (50 HP)	200	200	Shifting of Generator Requires as per PMC Design
		1 No x 4 Cusec (50 HP)			
		1 No x 6 Cusec (50 HP)			
		1 No x 10 Cusec (75 HP)			
3	Bird Market	1 No x 10 Cusec (100 HP)	400	300	Shifting of Generator Requires as per PMC Design
		1 No x 10 Cusec (100 HP)			
		1 No x 25 Cusec (200 HP)			
4	Larech Colony	1 No x 2 Cusec (20 HP)	100	200	Shifting of Generator Requires as per PMC Design
		1 No x 6 Cusec (60 HP)			
5	Ghazi Mohalla	1 No x 4 Cusec (30 HP)	100	200	Shifting of Generator Requires as per PMC Design
		1 No x 6 Cusec (60 HP)			
6	Muhammad Nagar	1 No x 6 Cusec (80 HP)	50	200	
7	Governor House	1 No x 6 Cusec (60 HP)	2 x 200 KVA	Not Available	
		1 No x 6 Cusec (60 HP)			
		1 No x 6 Cusec (60 HP)			
8	Lakshmi Chowk	1 No x 25 Cusec (215 HP)	400 & 1000KVA	500 KVA + 500 KVA + 1000 KVA	Shifting of Generator Requires as per PMC Design
		1 No x 25 Cusec (215 HP)			
		1 No x 25 Cusec (215 HP)			
		1 No x 25 Cusec (215 HP)			
		1 No x 25 Cusec (215 HP)			
		1 No x 25 Cusec (215 HP)			
9	Lytton Road	1 No x 6 Cusec (60 HP)	200 KVA	200 KVA	Shifting of Generator Requires as per PMC Design
		1 No x 10 Cusec (80 HP)			
		1 No x 6 Cusec (40 HP)			
10	PIA Planterium	1 No x 25 Cusec (215 HP)	400 & 200 KVA	200 KVA & 1000 KVA	
		1 No x 12 Cusec (100 HP)			
		1 No x 12 Cusec (100 HP)			
		1 No x 12 Cusec (100 HP)			
		1 No x 25 Cusec (215 HP)			
11	Gurumangat Road	1 No x 25 Cusec (150 HP)	200 KVA	500 KVA	
		1 No x 15 Cusec (150 HP)			
		1 No x 15 Cusec (150 HP)			
12	Q Block Gulberg	1 No x 25 Cusec (150 HP)	200 KVA	200 KVA	
		1 No x 6 Cusec (50 HP)			
13	G Block Gulberg	1 No x 25 Cusec (150 HP)	1000 KVA	500 KVA	Shifting of Generator Requires as per PMC Design
		1 No x 25 Cusec (150 HP)			
		1 No x 25 Cusec (150 HP)			
		1 No x 25 Cusec (150 HP)			
		1 No x 25 Cusec (150 HP)			
14	Zafar Ali Road	1 No x 6 Cusec (50 HP)	100 KVA	200 KVA	Shifting of Generator Requires as per PMC Design
		1 No x 6 Cusec (40 HP)			
15	Shareef Colony	1 No x 6 Cusec (40 HP)	50 KVA	200 KVA	Shifting of Generator Requires as per PMC Design
		1 No x 6 Cusec (40 HP)			
16	Shadman Colony	1 No x 6 Cusec (40 HP)	2 x 630 KVA	630 KVA	
		1 No x 25 Cusec (150 HP)			
		1 No x 25 Cusec (150 HP)			
17	Shama	1 No x 10 Cusec (75 HP)	100 KVA	200 KVA	Shifting of Generator Requires as per PMC Design
		1 No x 6 Cusec (80 HP)			
		1 No x 6 Cusec (60 HP)			
18	Ichra	1 No x 6 Cusec (60 HP)	800 KVA	300 KVA	
		1 No x 6 Cusec (80 HP)			
		1 No x 6 Cusec (80 HP)			
19	Rasool Park	1 No x 25 Cusec (150 HP)	200 KVA & 630 KVA	500 KVA	Shifting of Generator Requires as per PMC Design
		1 No x 15 Cusec (110 HP)			
		1 No x 08 Cusec (80 HP)			
		1 No x 06 Cusec (60 HP)			
		1 No x 06 Cusec (110 HP)			
20	Poonch Road	1 No x 10 Cusec (80 HP)	400 KVA	400 KVA	Shifting of Generator Requires as per PMC Design
		1 No x 10 Cusec (80 HP)			
		1 No x 20 Cusec (150 HP)			
21	Jhan ki Devi	1 No x 01 Cusec (20 HP)	50 KVA	Not Available	
22	Centre Point	1 No x 08 Cusec (50 HP)	400 KVA	300 KVA	Shifting of Generator Requires as per PMC Design
		1 No x 06 Cusec (50 HP)			
		1 No x 06 Cusec (60 HP)			
23	Harbanspura /Karachi Phatak Disposal	5 No x 25 Cusec (150 HP)	1000KVA	1000KVA	Under Construction (Expected to be completed by 1st Quarter of Year 2026)

Annexure-D

